Integration by Parts

Exam code: 9MA0

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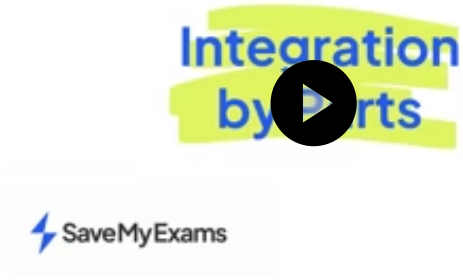

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Integration by parts

What is integration by parts?

INTEGRATION BY PARTS

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$$

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- For integrating the product of two functions - reverse product rule
- Crucially the product is made from **u** and **dv/dx** (rather than **u** and **v**)
- Alternative notation may be used

ALTERNATIVE NOTATION

FUNCTION NOTATION

$$\int f(x)g'(x)\,dx = f(x)g(x) - \int g(x)f'(x)\,dx$$

DASH NOTATION

$$\int uv'\,dx = uv - \int vu'\,dx$$

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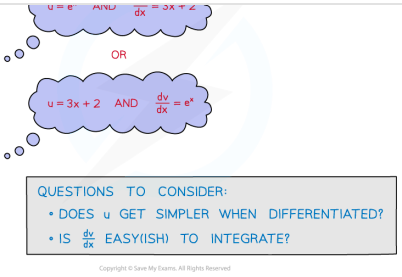
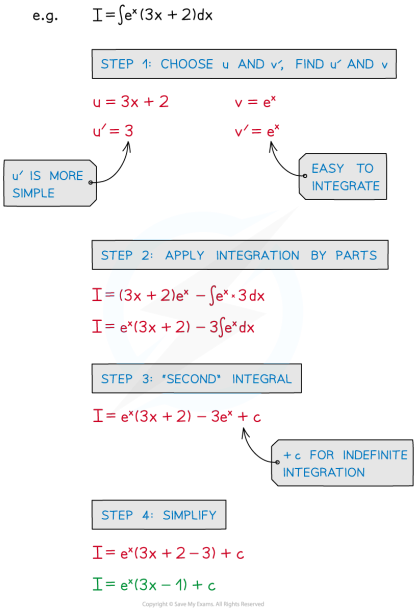
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

How do I use integration by parts?



- The hardest part is choosing **u** and **dv/dx** as there is no method for doing so
- **u**, ideally, becomes **simpler** when differentiated but this is not always possible
- **dv/dx** should be a function that can be integrated fairly easily
- Be wary of functions that 'cycle'/'repeat' when differentiated/integrated
  - $\mathbf{e^x \rightarrow e^x}$
  - $\mathbf{\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x}$



- **STEP 1**
  Choose u and v', find u' and v
- **STEP 2**
  Apply Integration by Parts
  - Simplify anything straightforward
- **STEP 3**
  Do the 'second' integral
  - If an **indefinite** integral remember "**+c**", the **constant of integration**
- **STEP 4**
  Simplify and/or apply limits

- It is possible that integration by parts may need to be applied more than once

e.g. $I = \int x^2 \cos x \, dx$

STEP 1: CHOOSE u AND v′, FIND u′ AND v

$u = x^2$ $\quad$ $v = \sin x$

$u' = 2x$ $\quad$ $v' = \cos x$

SIMPLER AFTER DIFFERENTIATING

"CYCLES" BETWEEN cos AND sin

STEP 2: APPLY INTEGRATION BY PARTS

$I = x^2 \sin x - \int 2x \sin x \, dx$

STEP 3: REPEAT STEP 1 AND STEP 2 FOR THE SECOND INTEGRAL

$u = 2x$ $\quad$ $v = -\cos x$

$u' = 2$ $\quad$ $v' = \sin x$

$I = x^2 \sin x - \left[-2x \cos x - \int -2 \cos x \, dx\right]$

STEP 4: INTEGRATE AND SIMPLIFY

$I = x^2 \sin x + 2x \cos x - 2 \sin x + c$

$I = (x^2 - 2) \sin x + 2x \cos x + c$

"+c"!!

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How do I integrate ln x?

$\int \ln x \, dx$

$\int \ln x \, dx = \int 1 \times \ln x \, dx$

$u = \ln x$ $\quad$ $v = x$

$u' = \frac{1}{x}$ $\quad$ $v' = 1$

$= x \ln x - \int x \times \frac{1}{x} dx$

$= x \ln x - \int dx$

$= x \ln x - x + c$

$\int \ln x \, dx = x \ln x - x + c$

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- A classic 'set piece' in almost every A level maths textbook ever written!
- In general, rewriting **f(x)** as **1×f(x)** can be a powerful problem-solving technique
- This could be a question in the exam

Can I find definite integrals using integration by parts?

- You can find the value of a definite integral using integration by parts
  - Use the layout shown in the example below

STEP 1: CHOOSE u AND v', FIND u' AND v

$u = 3x$ $\quad$ $v = \frac{1}{2}e^{2x}$ ← REVERSE CHAIN RULE

$u' = 3$ $\quad$ $v' = e^{2x}$

STEP 2: APPLY INTEGRATION BY PARTS

$$I = \left[\frac{3}{2}xe^{2x} - \int \frac{3}{2}e^{2x}\,dx\right]_0^1$$

$$I = \left[\frac{3}{2}xe^{2x} - \frac{3}{2} \times \frac{1}{2}\int 2e^{2x}\,dx\right]_0^1$$

STEP 3: DO THE "SECOND" INTEGRAL

$$I = \left[\frac{3}{2}xe^{2x} - \frac{3}{4}e^{2x}\right]_0^1$$

$$I = \left[\frac{3}{4}e^{2x}(2x - 1)\right]_0^1$$

STEP 4: APPLY LIMITS AND SIMPLIFY

$$I = \left(\frac{3}{4}e^{2}\right) - \left(\frac{3}{4}e^{0}x - 1\right)$$

$$I = \frac{3}{4}e^{2} + \frac{3}{4}$$

$$I = \frac{3}{4}(e^{2} + 1)$$

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Examiner Tips and Tricks

- Always think about what an elegant, slick, professional maths solution looks like – solutions normally get more complicated at first but quickly get simpler.
- If your work is continuing to get more complicated, stop and check for an error.
- Try to develop a sense of 'having gone too far down the wrong path'.
- This general advice is useful to remember:
- Is the second integral harder than the first?
- Try swapping your choice of **u** and **dv/dx**
- It is rare to have to apply integration by parts more than twice

Worked Example

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